Theory

Gaussian Beam Optics

Tyler Celotta
UAF

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Abstract

The purpose of the document is to work through the theory and experimental design for optimizing second-harmonic generation (SHG) at 1064 nm using a periodically poled lithium niobate (PPLN) crystal (50 × 2 × 0.5 mm³). In this document, I define Gaussian beam parameters, including beam waist, Rayleigh range, divergence, and confocal parameter, in order to determine the optimal focusing geometry. The Boyd-Kleinman condition for maximum CW SHG efficiency ($L/b = 2.84$) is a target focused waist of approximately 54.6 µm, but experimental constraints redefines a target of 77–80 µm ($L/b \approx 1.3 - 1.4$). Starting from a collimated 0.56 mm beam at 1064 nm with a measured divergence of 1.21 mRad, the collimator waist is estimated to be $\omega_{01} \approx 280$ µm, located 6.2 mm behind the collimator face. That being said, the calculated waist radius of roughly 0.279 mm would result in a beam larger than the crystal itself can accommodate. To account for this, I used an ABCD ray-transfer matrix to propigate a beam, constructing two equations relating a chosen lens focal length f to the required lens position d_1 and image distance d_2, with magnification $\alpha \approx 0.286$.

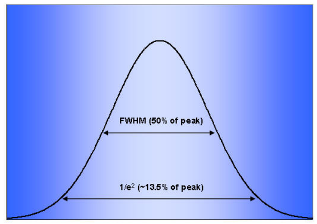
1 Experiment Specifics

1. Wavelength: 1064 nm
2. Collimator output beam size: 0.56 mm $\frac{1}{e^2}$ diameter
3. PPLN crystal: $50 \times 2 \times 0.5$ mm^3
4. Previous best measured output: 57 mW green, PPLN at 59.5°C
5. Previous calculated waist: about 45 μm
6. Desired next target: about 77–80 μm

2 Definitions and Equations

Beam Diameter ($\frac{1}{e^2}$): The $\frac{1}{e^2}$ beam diameter captures the diameter where the intensity of the beam drops to 13.53%; in other words, 86.47% of the beam intensity lies within the given diameter.

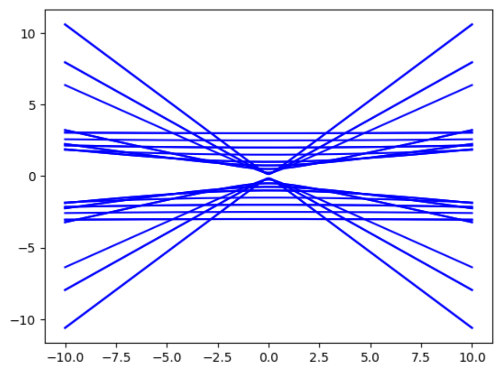
Key: ω_{01}: Initial Waist | ω_{02}: Focused Waist | z_{01}: Initial Rayleigh Range | α: Magnification | f: Focal Length
d_1: Dist. to Lens | d_2: Dist. to Focus



Beam Waist (ω_0): The beam waist is the portion of the beam where the radius is minimized, denoted ω_0. The tighter the focus the smaller the waist, but this introduces a large amount of divergence, as seen in the z-dependent beam width equation

$$\omega^2(z) = \omega_0^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_0^2}\right)^2\right] \tag{1}$$

Small waists are achieved by a "short focal length and a high numerical aperture largely filled by the input beam."[7]



Radius of Curvature (R_g): The radius of curvature is defined by Eq. 2. At its boundaries, $\lim_{z\to 0} R_g(z) = \infty$ where the wavefront is perfectly flat and acts like a section of a circle with infinite radius, and $\lim_{z\to\infty} R_g(z) = z$ where the wavefront averages out to a radius equal to the distance from the waist.

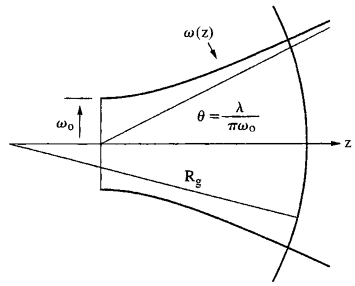
$$R_g(z) = z \left[1 + \left(\frac{\pi \omega_0^2}{\lambda z}\right)^2\right] \tag{2}$$

Rayleigh Range (z_0): The Rayleigh range is the distance at which the cross-sectional area of the beam is no more than twice that of the beam waist, ensuring at least 50% of the beam is retained.[8]

$$z_0 = \frac{\pi \omega_0^2}{\lambda} \tag{3}$$

Beam Divergence (θ): The beam divergence is the slope of the linear component of beam expansion (the far-field expansion), measured as a half-angle dependent on the wavelength and inversely on the waist:

$$\theta = \frac{\lambda}{\pi \omega_0} \tag{4}$$



Confocal Parameter (b): The confocal parameter uses the Rayleigh range to quantify the useful depth of focus—it is "the total length of the focused region around the beam waist."[8]

Boyd–Kleinman Focusing: $h = 1.068$ at $L/b = 2.84$ [1]; $h = 1.066$ at $L/b = 2.81$ [4]. Here h is the conversion efficiency, L is the crystal length, and $b = 2z_0$ is the confocal parameter. We work backwards from these values to find the optimal waist; see Section 3.2.

1

2

3 Calculations

3.1 Initial Beam Parameters

Starting from a 0.56 mm $1/e^2$ beam diameter at 1064 nm, what are the approximate Gaussian beam parameters?

Using Eq. 3:[3]

$$z_{01} = \frac{\pi\omega_{01}^2}{\lambda} = \frac{\pi \cdot (0.00028)^2}{1.064 \times 10^{-6}} \approx 0.23149 \text{ m} \approx 23.15 \text{ cm}$$

This results in $L/b = 0.108$, far from the Boyd–Kleinman optimum.

Using Eq. 4:

$$\theta = \frac{\lambda}{\pi\omega_{01}} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00028} \approx 1.210 \times 10^{-3} \text{ rad}$$

3.2 Boyd–Kleinman Focusing Targets

3.2.1 Optimum: $L/b = 2.84$ [1]

Given $L/b = 2.84$ and $L = 50$ mm.

$$b = \frac{L}{2.84} = \frac{0.05}{2.84} \text{ m}$$

Using Eq. 3:

$$\frac{0.05}{2.84} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \quad \Rightarrow \quad \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.84}} \approx 54.602\ \mu\text{m}$$

3.2.2 Optimum: $L/b = 2.81$ [4]

Given $L/b = 2.81$ and $L = 50$ mm.

$$b = \frac{L}{2.81} = \frac{0.05}{2.81} \text{ m}$$

Using Eq. 3:

$$\frac{0.05}{2.81} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \quad \Rightarrow \quad \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.81}} \approx 54.892\ \mu\text{m}$$

3.2.3 Conservative rule of thumb: $L/b = 1$ [2]

4

Given $L/b = 1$ and $L = 50$ mm.

$$b = L = 0.05 \text{ m}$$

Using Eq. 3:

$$0.05 = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \quad \Rightarrow \quad \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi}} \approx 92.017\ \mu\text{m}$$

3.2.4 Working Backwards from Target Waists

To recover a desired ω_{02}, we solve for $\xi \equiv L/b$:

1. $\omega_{02} = 45\ \mu$m:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (45 \times 10^{-6})^2} \approx 4.181$$

   This ratio is too large (waist too tight), leading to excessive diffraction and reduced efficiency.

2. $\omega_{02} = 77\ \mu$m:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (77 \times 10^{-6})^2} \approx 1.428$$

   Within the proper range.

3. $\omega_{02} = 80\ \mu$m:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (80 \times 10^{-6})^2} \approx 1.323$$

   Within the proper range.

3.3 Collimator Waist Estimation

Repeating the analysis without assuming the beam waist is at the collimator face.[5]

1. Using Eq. 4 to recover the waist from the measured divergence:

$$\omega_{01} = \frac{\lambda}{\pi\theta} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00121} \approx 279.9\ \mu\text{m}$$

2. Using the beam propagation equation (Eq. 5, from Eq. 2-27 in [5]):

$$\omega^2(z) = \omega_{01}^2 \left[1 + \left(\frac{\lambda z}{\pi\omega_{01}^2}\right)^2\right] \tag{5}$$

Solving for z:

$$\frac{\omega^2(z)}{\omega_{01}^2} - 1 = \left(\frac{\lambda z}{\pi\omega_{01}^2}\right)^2 \quad \Rightarrow \quad z = \pm\left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1}\right)\frac{\pi\omega_{01}^2}{\lambda}$$

Setting $\omega(z) = 0.00028$ m (the $1/e^2$ radius at the collimator aperture):

$$z \approx \pm\left(\sqrt{\frac{(0.00028)^2}{(0.0002799)^2} - 1}\right)\frac{\pi\cdot(0.0002799)^2}{1.064\times10^{-6}} \approx \pm 6.184 \text{ mm}$$

The waist is located approximately 6.2 mm *behind* the collimator face.[6]

3. Recalculating the Rayleigh range with the corrected waist:

$$z_{01} = \frac{\pi\omega_{01}^2}{\lambda} = \frac{\pi\cdot(2.799\times10^{-4})^2}{1.064\times10^{-6}} \approx 23.132 \text{ cm}$$

4. Because the beam waist $\omega_{01} \approx 280$ μm is the *minimum* radius of the laser, the beam can only diverge from that point and would "scrape" the inside of the PPLN crystal. A focusing lens is therefore required.

3.4 Lens Selection

1. The magnification relation from [3]:

$$\omega_{02} = \alpha\cdot\omega_{01} \quad \Rightarrow \quad \alpha = \frac{\omega_{02}}{\omega_{01}} \tag{6}$$

2. The lens-placement formula from [3]:

$$\alpha = \frac{f}{\sqrt{(|d_1| - f)^2 + z_{01}^2}} \tag{7}$$

Solving for $|d_1|$:

$$\alpha^2\left[(|d_1| - f)^2 + z_{01}^2\right] = f^2 \quad \Rightarrow \quad |d_1| = \sqrt{\left(\frac{f}{\alpha}\right)^2 - z_{01}^2} + f$$

3. Image-distance formula from [3]:

$$d_2 = f + \alpha^2(d_1 - f) \tag{8}$$

4. **Numerical values.** With $\omega_{02} = 80$ μm and $\omega_{01} = 279.9$ μm:

$$\alpha = \frac{80}{279.9} \approx 0.28582, \qquad z_{01} \approx 0.23132 \text{ m}$$

For a focal length f chosen in the lab:

$$d_1 = \sqrt{\left(\frac{f}{0.28582}\right)^2 - (0.23132)^2} + f, \qquad d_2 = f + (0.28582)^2(d_1 - f)$$

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5. **Derivation of equations used above** Starting from [5]:

$$\frac{1}{q_1} = \frac{1}{R_g} - i\frac{\lambda}{\pi\omega_{01}^2} \tag{9}$$

where R_g is the radius of curvature of the input beam. Taking the limit $R_g \to \infty$ (collimated input):

$$q_1 = i\frac{\pi\omega_{01}^2}{\lambda} = iz_{01}$$

The ABCD ray-transfer matrix for free-propagation d_1, thin lens f, then free-propagation d_2 is:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & d_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} 1 & d_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{pmatrix}$$

so $A = 1 - \frac{d_2}{f}$, $B = d_1 + d_2 - \frac{d_1 d_2}{f}$, $C = -\frac{1}{f}$, $D = 1 - \frac{d_1}{f}$.

The q-parameter transforms as (Eq. 10 from [5]):

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \tag{10}$$

Substituting $q_1 = iz_{01}$:

$$q_2 = \frac{\left(1 - \frac{d_2}{f}\right) iz_{01} + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)}{\left(-\frac{1}{f}\right) iz_{01} + \left(1 - \frac{d_1}{f}\right)}$$

Multiplying numerator and denominator by the complex conjugate of the denominator, $\left(1 - \frac{d_1}{f}\right) + i\left(\frac{z_{01}}{f}\right)$:

Real part:

$$\begin{aligned} \mathrm{Re}(q_2) &= \frac{-\left(1 - \frac{d_2}{f}\right)\left(\frac{z_{01}^2}{f}\right) + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)\left(1 - \frac{d_1}{f}\right)}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= \frac{d_2\left[\left(1 - \frac{d_1}{f}\right)^2 + \frac{z_{01}^2}{f^2}\right] + d_1\left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= d_2 + \frac{d_1\left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \end{aligned}$$

Imaginary part:

$$\mathrm{Im}(q_2) = \frac{z_{01}\left[\left(1 - \frac{d_1}{f} - \frac{d_2}{f} + \frac{d_1 d_2}{f^2}\right) + \left(\frac{d_1}{f} + \frac{d_2}{f} - \frac{d_1 d_2}{f^2}\right)\right]}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} = \frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2}$$

Full expression:

$$q_2 = \left[d_2 + \frac{d_1\left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right] + i \left[\frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right]$$

Setting the real part to zero to satisfy the waist condition: $\lim\limits_{z \to 0} \frac{1}{R_g(z)} = 0$ results in

$$d_2 = \frac{f\left[(d_1 - f)^2 + z_{01}^2\right] + f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2} = f + \frac{f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2}$$

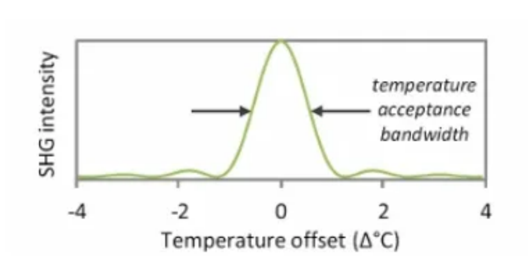
$$d_2 = f + \alpha^2(d_1 - f)$$

proving Eq. 8, where $\alpha^2 = \frac{f^2}{(d_1 - f)^2 + z_{01}^2}$ follows Eq. 7.

4 Use Case

PPLN:[7] To get the most out of our PPLN crystals, four key aspects must be considered:

1. **Crystal Length**
   (a) No special action required at this stage.
2. **Polarization**
   (a) The input light must be e-polarized, i.e. the polarization must be aligned with the dipole moment of the crystal. This is accomplished by aligning the polarization axis of the light parallel to the thickness of the crystal.[2]
3. **Focusing and the Optical Arrangement**
   (a) For SHG with CW lasers, the Boyd–Kleinman result shows that optimum conversion efficiency is achieved when $L/b = 2.84$.[2]
4. **Temperature and Period**
   (a) The crystal is manufactured with periodic stripes of alternating polarity with a spacing called the poling period. This quasi-phase-matched arrangement allows the harmonic power to build up continuously.
   (b) The effective poling period can be tuned by changing the crystal temperature, which causes thermal expansion or contraction.
   (c) Note: The refractive index of the crystal also changes with temperature.
   (d) The output intensity follows a sinc^2 curve (shown below), defining the temperature acceptance bandwidth.



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(e) Acceptance bandwidth is inversely proportional to crystal length.

(f) The optimum temperature can be found by heating the crystal to 20°C above the calculated phase-matching temperature, then allowing it to cool while monitoring the generated output power.[2]

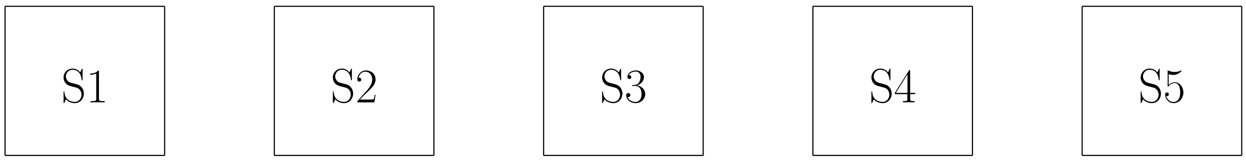


Figure 1: S1: Laser at 1064 nm. S2: [Component — fill in]. S3: [Component — fill in]. S4: [Component — fill in]. S5: [Component — fill in].

Notes

[1]Wikipedia entry: Second-harmonic generation (SHG), also known as frequency doubling, is the lowest-order wave-wave nonlinear interaction that occurs in various systems, including optical, radio, atmospheric, and magnetohydrodynamic systems. As a prototype behavior of waves, SHG is widely used, for example, in doubling laser frequencies. SHG was initially discovered as a nonlinear optical process in which two photons with the same frequency interact with a nonlinear material, are "combined", and generate a new photon with twice the energy of the initial photons (equivalently, twice the frequency and half the wavelength), conserving the coherence of the excitation. It is a special case of two-photon sum-frequency generation, and more generally of harmonic generation.

[2]**Phase-matching:** For sum-frequency generation to occur efficiently, phase-matching conditions must be satisfied: $\hbar k_3 \approx \hbar k_1 + \hbar k_2$, where k_1, k_2, k_3 are the angular wavenumbers of the three waves travelling through the medium. (This equation resembles conservation of momentum.) The sum-frequency generation becomes more efficient as this condition is satisfied more precisely.

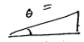
[3]In this calculation I assume the full beam diameter $2\omega_{01} = 0.56$ mm because the beam emerges from a collimator. This assumes the beam waist may be located a short distance away from the collimator face, as noted in [6]: "The purpose of a beam collimator is essentially to transform a strongly diverging light beam into a collimated beam, i.e., a beam where light propagates essentially only in one direction, and the beam divergence is weak. The output beam may have its beam waist close to the output aperture, or a mild focus somewhat away from it."

[4]"In general, a good rule of thumb is that the spot size should be chosen such that the Rayleigh range is half the length of the crystal." If $z_{02} = L/2$, then $b = 2z_{02} = L$, so $L/b = 1$.

[5]Note: 1.21 mRad was a measured value, substantially lower than the spec sheet value of approximately 5 mRad.

[6]The following calculations were made under the (incorrect) assumption of a waist radius of 45 μm rather than the updated 279.9 μm derived here.

$\theta =$



.25 mm

20 cm

$\theta = 1.25$ mrad

$\left.\begin{array}{l} 0.25 \text{ mm} \\ 0.3 \text{ mm} \\ 0.18 \text{ mm} \end{array}\right\}$ 0.243 mm

θ_B = Average divergence = 1.215 mrad

$\Rightarrow \omega_0 = 279$ um

For $\omega = 2.4$ mm$/2 = 1.2$ mm, $z = 76$ cm

$\Rightarrow$ the waist is 3.5 cm behind the collimator output

[7]Periodically poled lithium niobate (PPLN) is a lithium niobate crystal that switches polarity in a periodic fashion and is used for achieving quasi-phase-matching in nonlinear optics.

References

[1] G. D. Boyd and D. A. Kleinman. Parametric interaction of focused gaussian light beams. *Journal of Applied Physics*, 39(8):3597–3639, 1968.

[2] Covesion Ltd. Technical guide: How to use ppln. Covesion Knowledge Hub, 2026. Accessed: 2026-06-02.

[3] Edmund Optics. Gaussian beam propagation, 2026.

[4] HC Photonics. Optimal second harmonic generation. Technical document, HC Photonics, August 2017. Accessed: June 10, 2026.

[5] Gregory R. Osche. *Optical Detection Theory for Laser Applications*. John Wiley & Sons, 2002.

[6] Rüdiger Paschotta. Beam collimators. RP Photonics Encyclopedia. Accessed: 2026-06-02.

[7] Rüdiger Paschotta. Beam waist. RP Photonics Encyclopedia. Accessed: June 10, 2026.

[8] Rüdiger Paschotta. Rayleigh length. *RP Photonics Encyclopedia*, 2005.

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